Answers for the judges' questions

Simulated data. Every figure comes from the case's simulated data pack, not real Costco data, and is read from outputs/ when this sheet is built. Judging order: service, then labor cost, then feasibility. Answer in a breath, then offer the backup slide or the app page.

95.6%	+14.7%	87.0% / 89.5%	0
Expected service, BASE	**Labor cost, BASE: $377,504**	**Expected service, PEAK / DISRUPT**	**Blocks over capacity, DISRUPT**
Draft 68.5%	Draft $329,108	Draft 58.3% / 66.6%	Draft 12

1 The plan

What did the draft get wrong? F-19 · F-26 · F-48	It rolls recent habits forward: 99% of its blocks are within one person of the weeks 3-6 average, and it's the same schedule in all three scenarios. So hours sit in the wrong places: **135 of 180** BASE blocks fall below the 95% goal, for **68.5%** expected service.
Why aim for a 95% chance in each block? F-49 · A-13	Service is judged first, and 95% is where each extra point starts to get expensive. In BASE, a point of expected service costs about **$3,400** between the 85% and 90% goals, **$6,500** between the 92.5% and 95% goals, and **$24,800** between the 97.5% and 99% goals.
Isn't +14.7% labor cost a lot? F-51 · F-48	Moving hours alone, at about the draft's cost (**$328,220** against $329,108), already beats the draft in every condition we tested. The other **$49,284** buys allowances for forecast error and callouts, which lift expected service from 73.4% to **95.6%**.
What if the judges don't apply callouts? A-03 · F-04 · F-51 · Q-02	Then the callout allowance costs **$22,743** for little gain in their score: 99.6% expected service against 98.3% without it. We planned for real conditions, because **5.2%** of scheduled staff called out in history and only 49% were replaced.
Is the plan actually feasible? F-67 · F-70 · F-71	Yes. All **540** open blocks across the three scenarios are at or above minimum staff and within capacity under the README's staffing rule, and the 27 closed blocks are 0. An independent rebuild of the 567-row file finds no problems with the staffing.
Why did you cut staff in some blocks? F-39 · F-72	History shows some blocks are habitually overstaffed: **3,184 hours** sat above need, worth $103,458, while 2,472 were missing elsewhere. In BASE the plan cuts **12** blocks and adds to 132, and each says why: Warehouse 4 Merchandising's Wed Feb 18 opening goes from 21 to 18 and keeps a **98%** chance.
Why are some blocks still below the goal? F-50 · F-55	Capacity runs out: every block below the goal already uses all its own regular staff and overtime. In BASE, **22 of the 25** fall on the holiday weekend, Feb 13-16, and they still average an **83%** chance, with 11 at 90-95%. Limiting holiday PTO is the most direct fix.
Why did expected service in PEAK only reach 87%? F-50 · F-10 · F-48	PEAK raises demand but not the staff available, so capacity runs out sooner: the plan is **190 people short** of need across 45 blocks, 113 of them on Feb 14. It's still **28.7 points** above the draft's 58.3%, and the playbook says to arrange outside cover.
Why does DISRUPT cost less than BASE? F-42 · F-48 · F-71	Fewer people are available, so the plan fields about 370 fewer hours, at $366,649. The draft ignores the cuts: it schedules **36 more people than can be fielded** in 12 blocks, and the plan has **0** over capacity.

2 The forecast

How accurate is the forecast? F-44	On held-out weeks 4-6, daily error was **2.4%** against the benchmark's 4.1%. The edge comes from Friday to Monday, where the benchmark's average still carries earlier holiday days (2.4% against 5.1%); on normal planning-week days the two agree within 0.1%. So its value is that **it removes holiday distortion**; we don't claim it beats the benchmark in the planning week.
How do you know Presidents' Day behaves like MLK? F-45 · A-05 · A-11	We can't be sure, because MLK is the only holiday in the history with the same Friday-to-Monday shape. In a test, forecasting the MLK weekend with New Year's effect beat the benchmark for Front End (**8.2%** against 12.4%) and Merchandising (8.7% against 11.3%) but not Pharmacy, so the risk model adds extra uncertainty on Feb 13-16.
Why add 3% for Valentine's Day? F-52 · A-06	There's no Valentine's Day in the history, so +3% for Front End and Merchandising on Feb 14 is a stated judgment call. It barely matters: even if the true lift were 10%, planning for it would raise expected service there only from **64.5% to 65.9%**, because 11 of the 18 blocks are already at capacity.

What about traffic, promotions and weather? F-41 · F-61 · F-10	Promotions fall on every Friday to Sunday and holiday in the data, so their effect can't be separated, and the weekday and holiday terms carry it. Temperature adds nothing once the weekday is known (correlation **-0.006**), and the data pack says traffic is an outcome never to use as an input. PEAK's promotion surge comes in through the case's multipliers.

3 The model and its checks

Why not staff to 85% utilization, like the starter query? F-03 · F-17 · F-46	It sits on the edge of the cliff and ignores callouts. In history, blocks at 0.85-0.90 met target only **62%** of the time, and above 0.90 just 0.6%. Our limits for a 95% chance run from **0.837 to 0.868** by warehouse and department and apply to staff who actually work, so the plan adds room for callouts and forecast error.
How did you check the risk model? F-46 · F-62 to F-65 · D-29	Leaving one week out at a time, its predicted chances track real outcomes: **97.4% predicted against 95.7% observed** in the 95-99% group, and a Brier score of 0.024 against 0.177 for a flat rate. When a calibration study found it ran slightly high near the goal, we switched to a smooth curve that fits better on every measure, and it agrees closely with a check against history that uses no model.
How do you know the code is right? F-66 to F-72 · F-54	The full test suite passes, and rerunning the pipeline reproduces every output byte for byte. Separate checks that import none of our code rebuild the submission, the cost rule and the search from scratch and agree on every block, and a test confirms all **567** explanations quote their own block's numbers.
What if illness is worse than DISRUPT? F-69 · F-50 · Playbook	DISRUPT already cuts deeper than January's real flu week, and **18 of its 21** illness blocks are already at capacity, so no schedule can absorb more. The playbook says to escalate early for outside cover, and the Scenario lab cuts available staff further and replans live to show where the new gaps land.

4 The tool

Why React and FastAPI? D-24	The analysis stays in tested Python, and FastAPI serves it to a real web front end, the way a production tool would be built. It runs on one laptop and needs no internet connection, and with default settings it shows exactly the submitted plan.
Why totals per block instead of individual schedules? F-15 · Report ch. 5	The submission asks for total employees per block, and the data has no individual employee records, skills or preferences. Totals are the decision a planner makes first; the scheduling system that holds those records then turns them into named shifts.

5 Doing it for real

What's the single most useful thing a warehouse leader can do? F-55 · A-16	Limit holiday-weekend PTO approvals. Keeping up to 5 people per block off holiday PTO cuts BASE blocks below the goal from 25 to **6**, lifts expected service to **97.2%**, and saves $4,381 because regular staff replace overtime. It assumes some PTO is still open to change, which the data doesn't show.
How would Costco run this for real? Report ch. 10 · D-24	Feed it each week's demand, availability and callout data, rerun it, have warehouse leaders review the board and the blocks below the goal, then publish. The engine and the screens are already separated by an API, so a data feed and Costco's scheduling system could plug in without a rebuild.
What would you do with more time or data? A-05 · A-06 · A-16 · Q-02 · Q-03	Real outcomes from a Presidents' Day and a Valentine's Day, to replace the borrowed and judged effects; employee records, to turn totals into named shifts; and how much holiday PTO is already approved. We'd also ask how callouts are scored and how shared staff are counted, and track actual waits each week to refit the service curve.